

Vignesh M

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PROFESSIONAL SUMMARY

AI & Data Science graduate with strong expertise in **Software development** and a passion for emerging technologies. **Quick learner** and **collaborative team player** with experience in building scalable, user-focused applications. Motivated to deliver innovative solutions, contribute to organizational growth, and continuously enhance technical skills.

WORK EXPERIENCE

QWAT INNOVATION | FULL STACK INTERN | [LINK](#)

2 Months | Coimbatore, India

- Built a full-stack hospital management system using **Next.js**, **Tailwind CSS**, **Firebase Auth**, and **MySQL** modeled across **18 relational database tables**.
- Developed **4 role-based dashboards** (Admin, Receptionist, Pharmacist, Doctor) with secured role based access.
- Developed a real-time **pharmacy inventory assistant** to query medicine stock levels and automatically detect **low-stock medicines**.
- Tech Stack: **Next.js**, **Tailwind CSS**, **Firebase Auth**, and **MySQL**.

NXT LOGIC | AI ENGINEER INTERN | [LINK](#)

2 Months | Coimbatore, India

- Engineered a 2-stage hybrid RAG pipeline combining **FAISS semantic search** and **BM25 keyword search** with cross-encoder reranking.
- Generated **384-dimensional embeddings locally** with 500-character chunking (10% overlap), eliminating external API costs.
- Built 4 FastAPI endpoints supporting **3 LLM providers (Groq, OpenAI, Anthropic)**, and achieved a **100% pass rate on a 19-question evaluation suite**.
- Tech Stack: **Python** • **FastAPI** • **FAISS** • **BM25**.

PROJECTS

SMART ID VERIFICATION | [LINK](#)

Applied Computer Vision & Deep Learning

- Trained an **EfficientNet-B0 classifier** on 1,855 images across 3 ID types (Aadhaar, PAN, Passport), achieving **100% test accuracy**.
- Deployed 2 FastAPI endpoints with **image-quality filters** and **PaddleOCR parsers** extracting structured identity fields.
- Implemented data validation using **Verhoeff/MRZ checksums** and 512-dim face embedding matching via cosine similarity.
- Tech Stack: **Python**, **FastAPI**, **PyTorch**, **OpenCV**, **PaddleOCR**.

NETFLIX STUDIO | [LINK](#)

Full-Stack AI Streaming Web Application

- Integrated **Gemini 2.5 Flash** to build an AI-powered movie recommendation engine interpreting natural-language queries.
- Connected **TMDB REST APIs** for real-time catalogs and implemented **Firebase Authentication** with protected routes.
- Optimized frontend performance using **Redux Toolkit**, search debouncing, and API caching, reducing network calls by **>40%**.
- Tech Stack: **React.js**, **Node.js**, **Gemini API**, **Firebase**, **Redux**.

EDUCATION

SRI RAMAKRISHNA ENGINEERING COLLEGE

B.TECH AI & DATA SCIENCE
Oct 2022 – 2026 | Coimbatore

RJ MATRIC HR. SEC. SCHOOL

HSC

Grad. May 2022 | Coimbatore

SSLC

Grad. Mar 2020 | Coimbatore

LINKS

GitHub: [Vicky6804](#)

Portfolio: [vignesh.app](#)

TECHNICAL SKILLS

LANGUAGES

Advanced: JavaScript • Python • C++ • HTML • CSS • SQL

Intermediate: TypeScript • Java

FRAMEWORKS

React.js • Next.js • Node.js • Express.js • FastAPI • Tailwind CSS

AI / ML & VISION

RAG • FAISS • BM25 • PyTorch • OpenCV • PaddleOCR

TOOLS

Postman • MySQL • Firebase • GitHub • Vercel • VS Code

SOFT SKILLS

Critical Thinking • Teamwork

CERTIFICATIONS

Full Stack Web Development
-Udemy

AWS Cloud Foundation
-AWS Academy

Python Essentials

-Cisco Networking Academy

COURSEWORK

Data Structures and Algorithms
Object Oriented Programming
Database Management Systems
Machine Learning & Deep Learning
Operating Systems